SIMATS ENGINEERING
ASSESSMENT TOOL 1

COURSE NAME: CSA16 COURSE CODE: Data Warehousing and Data Mining

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COURSE OUTCOME COVERED

CO2: Apply suitable Data Pre processing techniques like Data cleaning,Data integration, Data transformation and data Reduction ,for effective data preparation (BL3,BL4)
Assessment Tool Weightage: 50%

QUESTIONS: 5 **Total Marks:50**

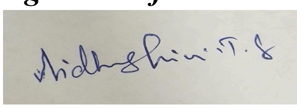
Annexure B
Analytical Problem Solving

Criteria	Problem Understanding ***(2)***	Method Selection ***(2)***	Solution Process ***(2.5)***	Accuracy of Calculation ***(2)***	Interpretation of Results ***(1.5)***	***Total(10)***
Weightage	*20%*	20%	25%	20%	15%	*100%*
Q1						
Q2						
Q3						
Q4						
Q5						

Code of Conduct:

I Vidhushini.T.S(192511061) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:



RUBRICS FOR CALCULATION:

Numerical Problems					
Criteria	**Weightage**	**Level 4 (Exemplary)**	**Level 3 (Proficient)**	**Level 2 (Developing)**	**Level 1 (Beginning)**
Problem Understanding	20%	Clearly interprets problem, identifies all parameters and requirements accurately	Understands problem with minor gaps	Partial understanding; misses key elements	Misinterprets or unable to understand problem
Method Selection	20%	Selects most appropriate method/formula with strong justification	Selects correct method with minor errors	Inappropriate or partially correct method	Unable to select method
Solution Process	25%	Logical, systematic steps with correct approach throughout	Mostly correct steps with minor mistakes	Disorganized steps; multiple errors	No clear process
Accuracy of Calculation	20%	All calculations accurate; correct final answer	Minor calculation errors	Frequent errors; incorrect final answer	No valid calculations
Interpretation of Results	15%	Clearly interprets results with units and engineering relevance	Basic interpretation with minor omissions	Limited or unclear interpretation	No interpretation

CO2 - Assessment Tool 1

1. Suppose that the data for analysis includes the attribute age. The age values for data tuples are (in increasing order) 13, 15, 16, 16, 19, 20, 20, 21, 22, 22, 25, 25, 25, 25, 30, 33, 33, 35, 35, 35, 35, 36, 40, 45, 46, 52, 70

a) What is the mean of the data? What is median?

$$\text{Mean} = \frac{1}{n}\sum_{i=1}^{n} x_i = \frac{1}{27} \times 809$$

$$= 30$$

$$\text{Median} = \frac{n+1}{2} = \frac{28}{2} = \underline{14} \Rightarrow 25$$

b) What is the mode of data? Comment on the data's modality (i.e, Bimodal, Trimodal, etc)

Solution:

25 and 35 are repeating for 4 times. So it is Bimodal.

c) What is midrange of the data?

$$\text{Midrange} = \frac{\max + \min}{2} = \frac{70 + 13}{2} = 41.5$$

d) Can you find the first quartile (Q1) and third quartile (Q3) of the data?

13, 15, 16, 16, 19, 20, 20, 21, 22, 22, 25, 25, 25

$$\frac{13+1}{2} = 7 \Rightarrow 20 \qquad Q_1 = 20$$

30, 33, 33, 35, 35, 35, 35, 36, 40, 45, 46, 52, 70

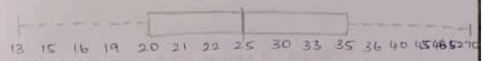
$$\frac{13+1}{2} = 7 \Rightarrow 35 \qquad Q_3 = 35$$

$$IQR = Q_3 - Q_1 = 35 - 20 = 15.$$

e) Give the five-number summary of the data

min, Q_1, median, Q_3, max – 13, 20, 25, 35, 70

f) Show a boxplot of the data.



2. Given the following measurements for the variable age: 18, 22, 25, 42, 28, 43, 33, 35, 56, 28. Standardize the variable by the following.

Compute the mean absolute deviation of age.

$$\underset{(\bar{x})}{\text{Mean}} = \frac{1}{n}\sum_{i=1}^{n} x_i = \frac{1}{10} \times 330 = 33.$$

Age	$\lvert x-\bar{x}\rvert$
18	$\lvert 18-33\rvert = 15$
22	$\lvert 22-33\rvert = 11$
25	$\lvert 25-33\rvert = 8$
42	$\lvert 42-33\rvert = 9$
28	$\lvert 28-33\rvert = 5$
43	$\lvert 43-33\rvert = 10$
33	$\lvert 33-33\rvert = 0$
35	$\lvert 35-33\rvert = 2$
56	$\lvert 56-33\rvert = 23$
28	$\lvert 28-33\rvert = 5$
	88

Mean Absolute Deviation:

$$= \frac{\sum \lvert x-\bar{x}\rvert}{n} = \frac{88}{10} = 8.8$$

3. Suppose that the data for analysis includes the attribute age. The age values for the data tuples are (in increasing order) 13, 15, 16, 16, 19, 20, 20, 21, 22, 22, 25, 25, 25, 25, 30, 33, 33, 35, 35, 35, 35, 36, 40, 45, 46, 52, 70. Use smoothing by bin means to smooth the above data, using a bin depth of 3. Illustrate your steps.

Total no. of values = 27

Total no. of bin depth = 3

$$\text{Total no. of Bins required} = \frac{27}{3} = 9$$

Bin	Values	Mean	Smoothed values
B1	13, 15, 16	14.67	14.67, 14.67, 14.67
B2	16, 19, 20	18.33	18.33, 18.33, 18.33
B3	20, 21, 22	21.00	21.00, 21.00, 21.00
B4	22, 25, 25	24.00	24.00, 24.00, 24.00
B5	25, 25, 30	26.67	26.67, 26.67, 26.67
B6	33, 33, 35	33.67	33.67, 33.67, 33.67
B7	35, 35, 35	35.00	35.00, 35.00, 35.00
B8	36, 40, 45	40.33	40.33, 40.33, 40.33
B9	46, 52, 70	56.00	56.00, 56.00, 56.00

4. Use the two methods below to normalize the following group of data: 200, 300, 400, 600, 1000.

(a) min-max normalization by setting max = 1, min = 0

min = 200 max = 1000

max − min = 1000 − 200 = 800

Original value	Calculation	Normalization
200	(200−200)/800 =	0
300	$(300-200)/800 = \frac{100}{800}$	0.125
400	$(400-200)/800 = \frac{200}{800}$	0.250
600	$(600-200)/800 = \frac{400}{800}$	0.500
1000	$(1000-200)/800 = \frac{800}{800}$	1.000

(b) Z-score normalization:

$$Z = \frac{x - \mu}{\sigma} \qquad \text{Mean} = \frac{2500}{5} = 500$$

Standard deviation:

x	$x-\mu$	$(x-\mu)^2$
200	200 − 500 = −300	90000
300	300 − 500 = −200	40000
400	400 − 500 = −100	10000
600	600 − 500 = 100	10000
1000	1000 − 500 = 500	250000
		400000

$$\sigma^2 = \frac{400000}{5}$$

$$\sigma^2 = 80000$$

$$\sigma = \sqrt{80000} \approx 282.84$$

Z-Scores

Original value	Calculation	Z-Score
200	(200-500)/282.84	-1.06
300	(300-500)/282.84	-0.71
400	(400-500)/282.84	-0.35
600	(600-500)/282.84	0.35
1000	(1000-500)/282.84	1.77

$\boxed{-1.06, -0.71, -0.35, 0.35, 1.77}$

5 Suppose a group of 12 sales price records has been sorted as follows: 5, 10, 11, 13, 15, 35, 50, 55, 72, 92, 204, 215. Partition them into three bins by each of the following methods. (a) equal-frequency partitioning (b) equal-width partitioning (c) clustering

Total no. of records = 12

Total no. of bins = 3

a) ~~Recor~~ Equal-Frequency Partitioning = $\frac{\text{Record}}{\text{bins}} = \frac{\text{Total}}{\text{no. of bins}}$

$= \frac{12}{3} = 4.$

Bin	Values
Bin 1	= 5, 10, 11, 13
Bin 2	= 15, 35, 50, 55
Bin 3	= 72, 92, 204, 215

b) Equal-Width Partitioning

$$\text{Width} = \frac{\text{Maximum} - \text{Minimum}}{\text{no. of Bins}}$$

$$= \frac{215-5}{3} = \frac{210}{3} = 70$$

Bin		Values
Bin 1	(5 to 74)	5, 10, 11, 13, 15, 35, 50, 55, 72
Bin 2	(75 to 144)	92
Bin 3	(145-215)	204, 215

c) Clustering

Cluster	Values
cluster 1	5, 10, 11, 13, 15
cluster 2	35, 50, 55, 72, 92
cluster 3	204 215

6 Suppose a hospital tested the age and body fat data for 18 randomly selected adults with the following result.

age	23	23	27	27	39	41	47	49	50	52	54	54	56	57	58	58	60	61
% fat	9.5	26.5	7.8	17.8	31.4	25.9	27.4	27.2	31.2	34.6	42.5	28.8	33.4	30.2	34.1	32.9	41.2	35.7

a) calculate mean, median and standard deviation for age and % fat.

Mean = $\frac{836}{18} = 46.44$

Mean = $\frac{518.1}{18} = 28.78$

Median

for age 51.00

for % fat = 30.70

$$\sigma^2 = \frac{1}{N}\sum_{i=1}^{n}(x-\mu)^2$$

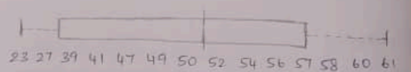
$$\sigma = \sqrt{\frac{1}{n}\sum_{i=1}^{n}(x_i-\mu)^2}$$

for age = 12.85 % fat = 8.99

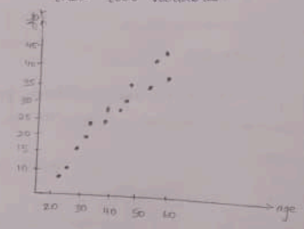
b) Draw the boxplots for age and % fat

min, Q_1, median, Q_3, max

23, 36.2, 51, 57, 61



c) Draw the scatter plot and a q-q plot based on these two variables



d) Normalize the two variables based on z-score normalization.

$$z = \frac{x-\mu}{\sigma} \qquad \mu = 46.44 \qquad \sigma = 12.85$$

Age	x-μ	z-score
23	-23.44	-1.82
23	-23.44	-1.82
27	-19.44	-1.51
27	-19.44	-1.51
39	-7.44	-0.58
41	-5.44	-0.42
47	0.56	0.04
49	2.56	0.20
50	3.56	0.28
52	5.56	0.43
54	7.56	0.59
54	7.56	0.59
56	9.56	0.74
57	10.56	0.82
58	11.56	0.90
58	11.56	0.90
60	13.56	1.06
61	14.56	1.13

e) Calculate the correlation coefficient. Are these two variables positively or negatively correlated

$$r = \frac{n\sum xy - \sum x \sum y}{\sqrt{(n\sum x^2 - (\sum x)^2)(n\sum y^2 - (\sum y)^2)}}$$

$\boxed{n = 18}$

x : age

y = % fat

$\boxed{\sum x = 836}$

$\boxed{\sum y = 518.1}$

$\boxed{(\sum x)^2 = 698896}$

$\boxed{(\sum y)^2 = 268427.61}$

$\boxed{\sum x^2 = 41,798}$

$\boxed{\sum y^2 = 16368.59}$

xy

218.5	1799.2
609.5	2295.0
210.6	1555.2
480.6	1870.4
1224.6	1721.4
1061.9	1908.2
1287.8	1977.8
1332.8	2472.0
1560.0	2177.7

$\boxed{\sum xy = 25763.2}$

$$r = \frac{18 \times 25763.2 - 433121.6}{\sqrt{(18 \times 41798 - 698896)(18 \times 16368.59 - 268427.61)}}$$

$$= \frac{468737.6 - 433121.6}{\sqrt{(752364 - 698896)(294634.62 - 268427.61)}}$$

$$= \frac{30606}{\sqrt{(53468)(262807.01)}}$$

$$= \frac{30606}{\sqrt{1401236411}} = \frac{30606}{37433.09}$$

$$= 0.817 \approx 0.82$$

As the value of person's product is 0.82 which is closer to 1 so the age and % fat are having ~~pos~~ strong positive correlation.